cross-platform software and can be installed in Linux, Windows and macOS systems. The command 'pip3 --version' shows the version of pip installed in your system, as shown below:

```
pip 20.0.2 from /usr/lib/python3/dist-packages/pip (python 3.8)
```

Now we need to install an integrated development environment (IDE) for Python. IDEs help programmers write, compile, debug and execute code very easily. There are many contenders for this position also. PyCharm, IDLE, Spyder, etc, are popular Python IDEs. However, since our primary aim is to develop AI and data science based applications, we consider two other heavy contenders — JupyterLab and Google Colab. Strictly speaking, they are not just IDEs; rather, they are very powerful Web based interactive development environments. Both work on Web browsers and offer immense power. JupyterLab is free and open source software supported by Project Jupyter, a non-profit organisation. Google Colab follows the freemium model, where the basic model is free and for any additional features a payment is required. I am of the opinion that Google Colab is more powerful and has more features than JupyterLab. However, the freemium model of Google Colab and my relative inexperience with it made me choose JupyterLab over Google Colab for this tutorial. But I strongly encourage you to get familiar with Google Colab at some point in your AI journey.

JupyterLab can be installed locally using the command 'pip3 install jupyterlab'. The command 'jupyter-lab' will execute JupyterLab in the default Web browser of your system. An older and similar Web based system called Jupyter Notebook is also provided by Project Jupyter. Jupyter Notebook can be locally installed with the command 'pip3 install notebook' and can be executed using the command 'jupyter notebook'. However, Jupyter Notebook is less powerful than JupyterLab, and

$$\begin{bmatrix} 11 & 22 & 33 \\ 44 & 55 & 66 \end{bmatrix}$$

Figure 2: A 2 x 3 matrix

it is now official that JupyterLab will eventually replace Jupyter Notebook. Hence, in this tutorial we will be using JupyterLab when the time comes. However, in the beginning stages of this tutorial we will be using the Linux terminal to run Python programs, and hence the immediate need for pip, the package management system.

Anaconda is a very popular distribution of Python and R programming languages for machine learning and data science applications. As potential AI engineers and data scientists, it is a good idea to get familiar with Anaconda also.

Now, we need to fix the most important aspect of this tutorial -- the style in which we will cover the topics. There are a large number of Python libraries to support the development of AI based applications. Some of them are NumPy, SciPy, Pandas, Matplotlib, Seaborn, TensorFlow, Keras, Scikit-learn, PyTorch, etc. Many of the textbooks and tutorials on AI, machine learning and data science are based on complete coverage of one or more of these packages. Though such coverage of the features of a particular package is effective, I have planned a more maths-oriented tutorial. We will first discuss a maths concept required for developing AI applications, and then follow the discussion by introducing the necessary Python basics and the details of the Python libraries required. Thus, unlike most other tutorials, we will revisit Python libraries again and again to explore the features necessary for the implementation of certain maths concepts. However, at times, I will also request you to learn some basic concepts of Python and mathematics on your own. That settles the final question about the nature of this tutorial.

After all this buildup, it would be a sin if we stop this article at this point without discussing even a single line of Python code or a mathematical object necessary for AI. Hence, we move on to learn one of the most important topics in mathematics required for conquering AI and machine learning.

Vectors and matrices

A matrix is a rectangular array of numbers, symbols or mathematical expressions arranged in rows and columns. Figure 2 shows a 2 x 3 (pronounced '2 by 3') matrix having 2 rows and 3 columns. If you are familiar with programming, this matrix can be represented as a two-dimensional array in many popular programming languages. A matrix with only one row is called a row vector and a matrix with only one column is called a column vector. The vector [11, 22, 33] is an example of a row vector.

But why are matrices and vectors so important in a discussion on AI and machine learning? Well, they are the core of linear algebra, a branch of mathematics. Linear algebraic techniques are heavily used in AI and machine learning. Mathematicians have studied the properties and applications of matrices and vectors for centuries. Mathematicians like Gauss, Euler, Leibniz, Cayley, Cramer, and Hamilton have a theorem named after them in the fields of linear algebra and matrix theory. Thus, over the years, a lot of techniques have been developed in linear algebra to analyse the properties of matrices and vectors.

Complex data can often easily be represented in the form of a vector or a matrix. Let us see a simple example. Consider a person working in the field of medical transcription. From the medical records of a person named P, details of the age, height in centimetres, weight in kilograms, systolic blood pressure, diastolic blood pressure and fasting blood sugar level in milligrams/decilitres can be obtained. Further, such information can easily be represented as the row vector.